

subject.
Physics

system.
Thermodynamics

Nitrogen is extremely cold in its liquid phase. It is used in the cryopreservation of small tissue samples and in cryosurgery to freeze and destroy diseased tissues. Liquid nitrogen can be produced from nitrogen gas using the Hampson-Linde cycle (Figure 1).

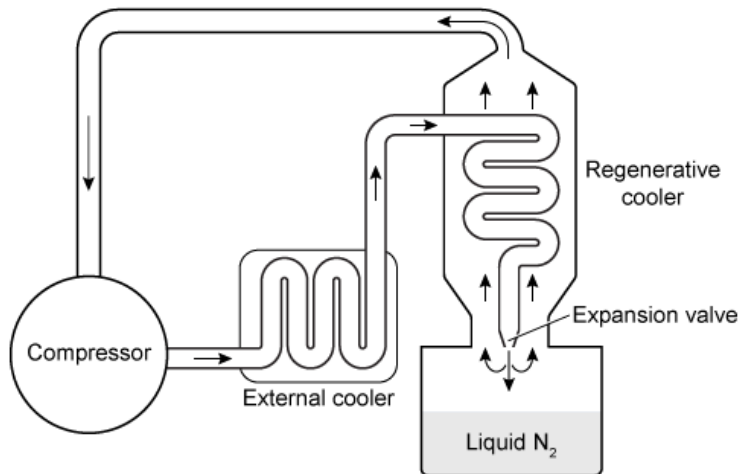


Figure 1 Hampson-Linde cycle apparatus used to liquefy nitrogen

The cycle begins by compressing nitrogen gas and forcing it through a coil in an external cooler that contains dry ice. The gas proceeds to a heat exchanger, where it cools further due to the countercurrent heat exchange with colder nitrogen gas returning from a later stage of the cycle; this process is known as regenerative cooling. The gas is subsequently forced through a narrow valve, where it expands and cools significantly upon exiting. As a result, a portion of the gas becomes liquefied.

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Table 1 Thermal Properties of Nitrogen

Boiling point at STP	77 K
Melting point at STP	63 K
Specific heat (liquid phase)	2 kJ/kg · K
Heat of vaporization	400 kJ/kg
Heat of fusion	50 kJ/kg

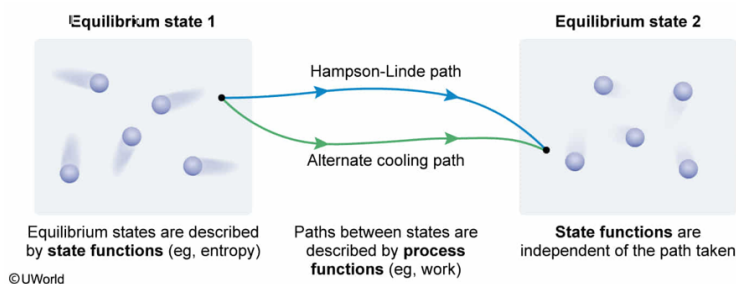
Which thermodynamic quantity involved in the Hampson-Linde cycle is a state function?

- ✓ ☒ A. Entropy of nitrogen, because it describes an equilibrium state of the system [65%]
- ☐ B. Entropy of nitrogen, because it describes a path to an equilibrium state of the system [14%]
- ☐ C. Work done by the compressor, because it describes an equilibrium state of the system [10%]
- ☐ D. Work done by the compressor, because it describes a path to an equilibrium state of the system [10%]

Correct

65% answered correctly

Explanation:



A system is in a state of **thermodynamic equilibrium** if the temperature of the system is constant and uniform throughout its volume and there is no flow of energy. A system is in thermodynamic equilibrium with another system or its surroundings if both have the same temperature.

State functions (or state quantities) describe the **equilibrium state** of a system as a relationship between various thermodynamic variables and are independent of the path taken by the system to arrive at its present state. State functions include a system's pressure, volume, and temperature.

Process functions (or path functions) describe the **path taken** by a system to transition from one equilibrium state to another. A system transitions from one state to another due to a net flow of energy in the form of heat transfer or work. For example, the loss or gain of heat is a process function because it describes the path taken by a system from its current pressure, volume, and temperature to a different set of values.

In the Hampson-Linde cycle described in the passage, work is done by the compressor on the nitrogen gas to decrease its volume. Therefore, work is a process function because it describes what was done to the system to change its state (**Choice C**). A system's **entropy** is a measure of its current **state of disorder** and does not depend on how the system arrived at that state (**Choice B**). Therefore, the entropy of nitrogen is a **state function**.

(**Choice D**) Although it is true that the work done by the compressor describes a path to an equilibrium state, work does not describe the state of the system. Therefore, work is a process function, not a state function.

Educational objective:

State functions describe the equilibrium state of a system as a relationship between various thermodynamic variables, and a process function describes the path taken by the system to go from one equilibrium state to another. Therefore, entropy is a state function, and work is a process function.

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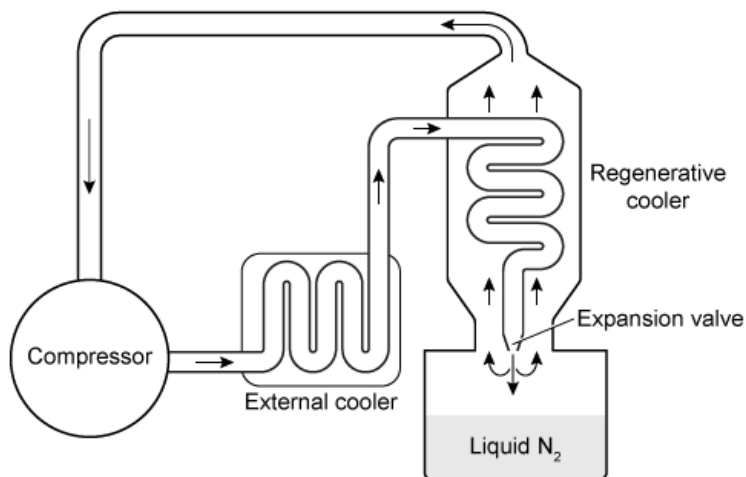


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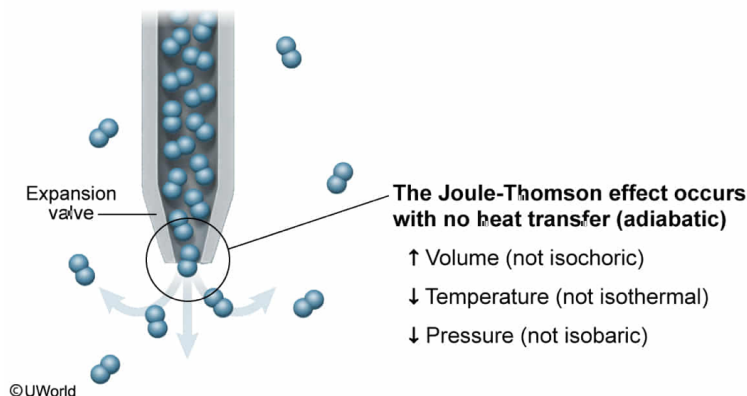
Based on the passage, a feature of the Joule-Thomson effect is that it occurs:

- ☐ A. isothermally. [32%]
- ✓ ☒ B. adiabatically. [56%]
- ☐ C. isochorically. [5%]
- ☐ D. isobarically. [5%]

Correct

57% answered correctly

Explanation:



Thermodynamic processes involve a change in one or more of the quantities describing the system. Processes in which an important thermodynamic quantity remains constant are assigned a special name. Four of these types of thermodynamic processes include:

- **Isothermal** processes, in which the temperature of the system does not change.

- **Adiabatic** processes, which occur with **no heat exchange** between the system and the environment.
- **Isochoric** (isovolumetric) processes, in which the volume of the system does not change.
- **Isobaric** processes, in which the pressure of the system does not change.

The Joule-Thomson effect is stated to occur **without heat transfer** with the surroundings. Therefore, the total heat content of nitrogen does not change, making it an **adiabatic** process.

(Choice A) The nitrogen cools (its temperature changes) as it exits the valve; therefore, the process is not isothermal.

(Choice C) The nitrogen gas expands (its volume changes) as it exits the valve; therefore, the process is not isochoric.

(Choice D) According to the ideal gas law ($PV = nRT$), the pressure of a gas is proportional to its temperature and inversely proportional to its volume. Because the nitrogen's temperature decreases (cooling) and volume increases (expansion), its pressure should be expected to decrease.

Educational objective:

Thermodynamic processes that keep important quantities constant are assigned special names. These include isothermal (constant temperature), adiabatic (without external heat transfer), isochoric (constant volume), and isobaric (constant pressure) processes.

Time spent:0 sec

Subject:
Physics

QID:401247

System:
Thermodynamics

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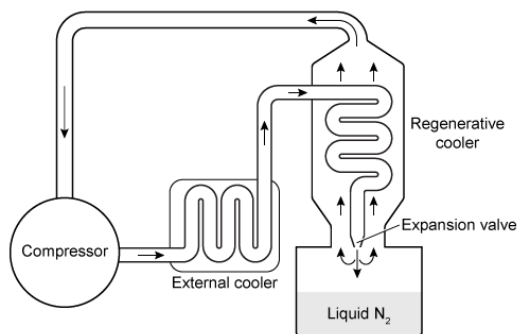


Figure 1 Hampson-Linde cycle apparatus used to liquefy nitrogen

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The compressor exerts a pressure of 400 Pa to compress nitrogen gas from 3 L to 1 L. If no heat transfer occurs, how much will the internal energy of the gas change? (Note: $1 \text{ m}^3 = 1000 \text{ L}$ and $1 \text{ J} = 1 \text{ Pa} \cdot \text{m}^3$.)

- ✓ ☒ A. 0.8 J [59%]
☐ B. 1.2 J [9%]
☐ C. 800 J [26%]
☐ D. 1200 J [4%]

Correct

59% answered correctly

Explanation:



The internal energy of the gas will increase by the work done

$$W = P\Delta V$$

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The internal energy of a system is the total energy stored in its molecules and is related to its temperature. Energy cannot be created nor destroyed; it can only transfer from one form to another. From the **First law of thermodynamics**, the change in internal energy (ΔU) is the sum of the amount of heat (Q) transferred to the system and the work (W) *done to the system* by the surroundings:

$$\Delta U = Q + W$$

In this case, nitrogen gas is considered the system and the compressor is considered the surroundings. **No direct heat transfer** occurs between the system and the surroundings ($Q = 0$). Despite this, the temperature of the gas increases because its internal energy increases by the amount of work done by the compressor:

$$\Delta U = W$$

The work associated with the expansion or contraction of a gas is known as **pressure-volume work**, which is calculated as the product of the external pressure (P) and the change in volume (ΔV):

$$W = P \Delta V$$

The work done by the compressor is positive because the compressor adds energy to the gas by compressing it from a volume of 3 L to 1 L ($\Delta V = 2\text{ L}$) at a pressure of 400 Pa. The unit for W is the J, which is equal to $1 \text{ Pa} \cdot \text{m}^3$. Hence, 2 L must be converted to cubic meters (m^3):

$$\Delta V = 2 \text{ L} \cdot \frac{1 \text{ m}^3}{10^3 \text{ L}} = 0.002 \text{ m}^3$$

Consequently, the work done on this gas is:

$$W = (400 \text{ Pa}) \cdot (0.002 \text{ m}^3)$$

$$W = 0.8 \text{ J}$$

Therefore, the internal energy of the gas increases by 0.8 J as the nitrogen gas is compressed from a volume of 3L to a volume of 1 L (**Choice A**).

Educational objective:

The internal energy of a system increases when heat is transferred to the system and when work is *done on the system* ($\Delta U = Q + W$). For a system of gases, the amount of work done is equal to the product of the external pressure and change in volume ($W = P\Delta V$).

Time spent: 0 sec
Subject:
Physics

QID: 401248
System:
Thermodynamics

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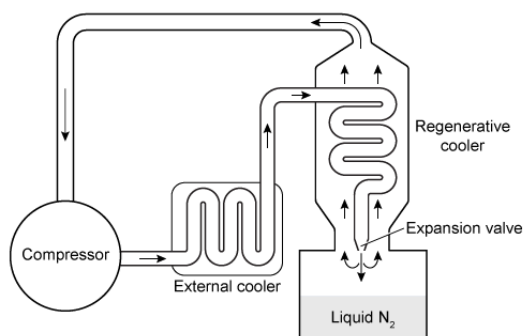


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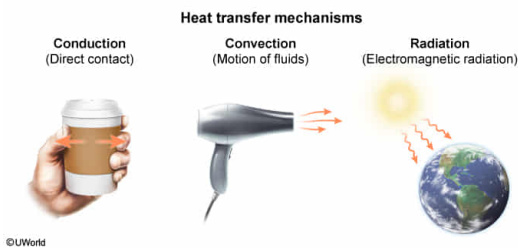
Which of the following best describes the type of heat transfer that occurs in the external cooler of the Hampson-Linde cycle apparatus?

- ☐ A. Conduction occurs as dry ice loses heat to the coil of the external cooler. [13%]
- ☐ B. Convection occurs as the coil of the external cooler loses heat to dry ice. [34%]
- ☒ C. Conduction occurs as the coil of the external cooler gains heat from nitrogen gas. [43%]
- ☐ D. Convection occurs as nitrogen gas gains heat from the coil of the external cooler. [8%]

Correct

43% answered correctly

Explanation:



Heat is the energy transferred from one object to another due to their temperature difference, and it always flows from hotter objects to colder objects. Heat can be transferred by three different mechanisms:

- **Conduction** is the transfer of heat through direct physical contact.
- **Convection** is the transfer of heat through the **flow of fluids**. Fluids absorb heat from hotter regions and deliver it to colder regions.
- **Radiation** is the transfer of heat through electromagnetic radiation, such as **infrared light**. Radiation heat transfer is significant only for high temperatures.

As the **nitrogen gas** flows through the coil of the external cooler, it transfers energy obtained during the compression stage to different sections of the coil. Therefore, the heat transfer results from the physical contact of the nitrogen gas and the coil and is primarily a **conductive transfer of heat**.

(Choice A) Because heat is transferred from hotter objects to colder objects, heat is *gained* (not lost) by the dry ice.

(Choice B) Both dry ice and the coil are solids. Therefore, these substances cannot participate in convective heat transfers.

(Choice D) Because heat is transferred from hotter objects to colder objects, heat is *lost* (not gained) by the nitrogen.

Educational objective:

Heat is energy that flows from high temperatures to low temperatures. Heat can be transferred through direct physical contact (conduction), through the flow of fluids (convection), or through emission of electromagnetic energy (radiation).

Time spent: 0 sec

Subject:
Physics

QID: 401249

System:
Thermodynamics

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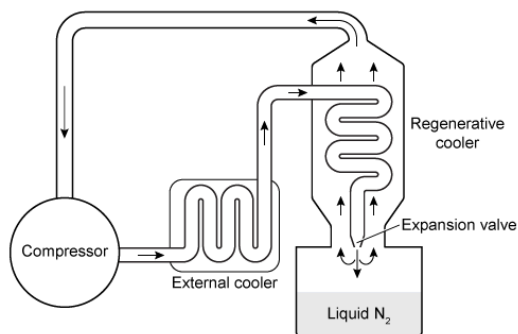


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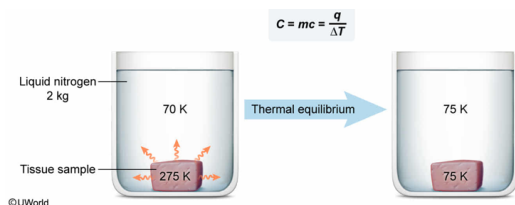
A tissue sample at 275 K is submerged in 2 kg of liquid nitrogen at 70 K for cryopreservation. The final temperature of the nitrogen is 75 K. What is the heat capacity of the sample in J/K? (Note: Assume no heat is lost to the surrounding air.)

- ☐ A. 0.05 J/K [11%]
- ☐ B. 0.1 J/K [23%]
- ☐ C. 50 J/K [32%]
- ✓ ☒ D. 100 J/K [32%]

Correct

33% answered correctly

Explanation:



Specific heat c is the amount of heat required to change the temperature of a unit mass of a material by 1 degree, and **heat capacity** C is the product of an object's mass m and its specific heat: $C = mc$. For a given heat q and a temperature change ΔT , the heat capacity of an object is:

$$C = mc = \frac{q}{\Delta T}$$

Rearranging this equation leads to the **calorimetry equation**:

$$q = C \Delta T = mc \Delta T$$

The given initial temperature of the liquid nitrogen is 70 K and the final temperature is 75 K; therefore, $\Delta T = 5$ K. The mass of nitrogen is given as $m = 2$ kg, and the specific heat of liquid nitrogen is given in Table 1 of the passage as $c = 2$ kJ/kg·K. Therefore, the amount of heat absorbed by nitrogen is:

$$q = (2 \text{ kg}) \left(2 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \right) (5 \text{ K}) = 20 \text{ kJ}$$

Because nitrogen reaches a final temperature at 75 K, the system must have reached **thermal equilibrium**. At equilibrium, temperature is uniform throughout the system, and therefore the final temperature of the tissue sample is also 75 K. Given that the tissue's initial temperature was 275 K, $\Delta T = -200$ K. The heat lost by the tissue was transferred to the nitrogen, $q = -20$ kJ. The heat capacity of the tissue is:

$$C = \frac{q}{\Delta T} = \frac{-20 \text{ kJ}}{-200 \text{ K}} = 0.1 \text{ kJ/K}$$

This value is converted to units of J/K using $1,000 \text{ J} = 1 \text{ kJ}$:

$$(0.1 \text{ kJ/K}) \left(\frac{1,000 \text{ J}}{1 \text{ kJ}} \right) = 100 \text{ J/K}$$

(Choices A and C) If the specific heat of nitrogen is incorrectly used as heat capacity, the tissue heat capacity is calculated to be 50 J/K, or 0.05 kJ/K.

(Choice B) The specific heat of nitrogen is given in kJ/kg·K. If the **kilo-** prefix is ignored, the incorrectly calculated heat capacity is 0.1 J/K.

Educational objective:

Heat capacity measures the amount of heat required to change the temperature of an object by 1

degree. Specific heat is an object's heat capacity per unit mass. The heat capacity C of an object can be calculated from its mass m and specific heat c ($C = mc$), or from the amount of heat q for a corresponding change in temperature ΔT : $q = mc\Delta T = C\Delta T$.

Time spent: 0 sec

Subject:
Physics

QID: 401250

System:
Thermodynamics

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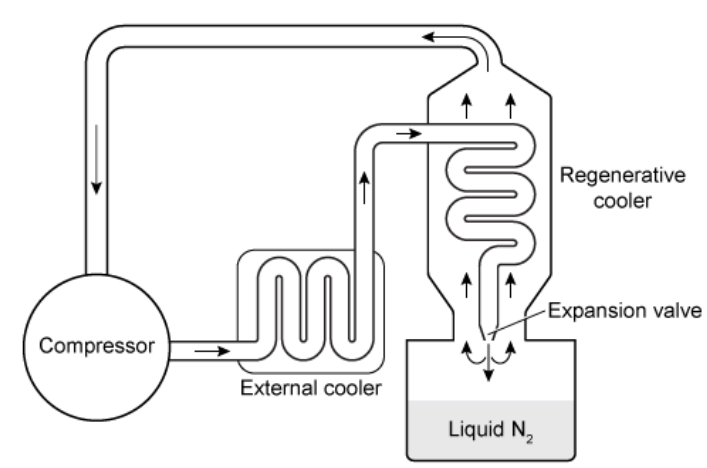


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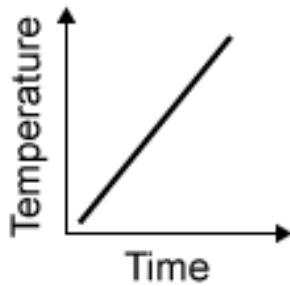
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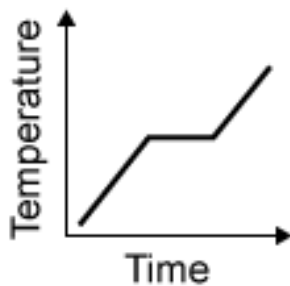
Heat of vaporization	400 kJ/kg
Heat of fusion	50 kJ/kg

Which of the following graphs best illustrates how the temperature of nitrogen changes as it is cooled from a gas and past its condensation point?

☐ A.



☐ B.



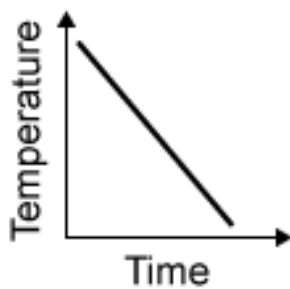
[6%]

✓ ☒ C.



[88%]

☐ D.



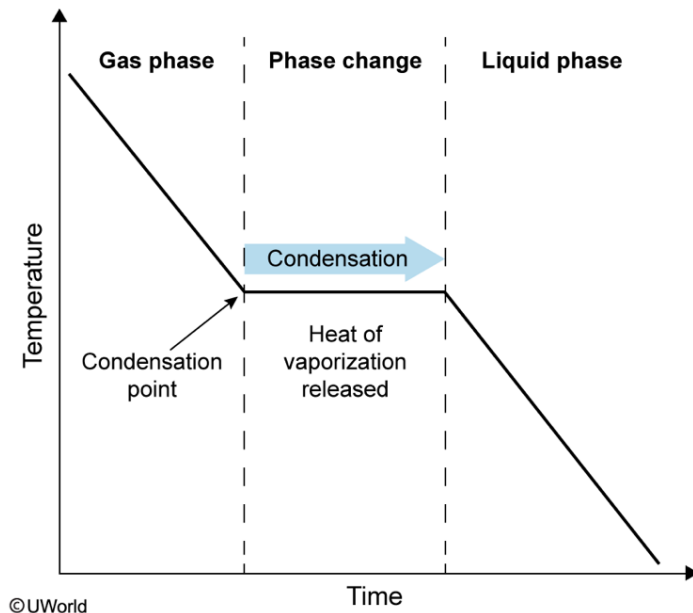
[4%]

Correct

88% answered correctly

Explanation:

Explanation:



Matter exists in three main phases: solid, liquid, and gas. When a substance changes from one phase to another (a **phase transition**), it absorbs (or releases) heat and its **temperature remains constant** until the entire substance has transitioned to the new phase. The condensation point is the temperature at which the phase transition from gas to liquid occurs, releasing heat equal to its **latent heat of vaporization**. Conversely, latent heat of vaporization is absorbed when a substance evaporates, which is the phase change from liquid to gas.

The question states that nitrogen is cooled from a gas and past its condensation point. Hence, heat is being taken away from nitrogen. Therefore, the temperature of nitrogen initially decreases until it reaches the condensation point. At the condensation point, temperature remains constant because the removal of heat causes the gas to condense into liquid. The temperature decreases again after all the nitrogen has completed the phase change.

(Choice A) Because the nitrogen was cooled, its temperature decreases as time progresses.

(Choice B) This is the **phase change diagram** of the process. A phase change diagram shows how the temperature of a substance changes as energy (heat) is added to it, resulting in a linearly increasing temperature, except during phase transitions, when the temperature remains constant.

(Choice D) If the latent heat of vaporization is incorrectly ignored, it results in the line graph with the negative slope shown.

Educational objective:

Matter exists in three main phases: solid, liquid, and gas. During a phase transition, the temperature of a substance remains constant. The condensation point is the temperature at which the phase transition from gas to liquid occurs, releasing heat equal to its latent heat of vaporization.

Time spent:0 sec

Subject:
Physics

QID:401251

System:
Thermodynamics